Krishna Sruthi Velaga

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Summary

First-year PhD student working on model compression and encoding techniques in federated edge learning. Previously, built instrumented testbeds and conducted empirical studies on quantized large language models for smart home automation, analyzing latency, energy consumption, and thermal characteristics.

Experience

Doctoral Research Assistant, Department of Computer and Information Sciences, Towson University Aug 2025 – Present

- Reviewed 241 papers on Edge AI, mapping applications, data sources, models, and infrastructure across five urban domains.
- Develop compression technique (pruning, quantization) in federated learning during local training on heterogeneous devices to reduce computation and communication overhead.
- Design lossless encoding techniques to exploit sparsity and reduce model transmission size with exact reconstruction.

Doctoral Teaching Assistant, Department of Computer and Information Sciences, Towson University Aug 2025 – Present

- Grade programming assignments for 60 students weekly, providing feedback on algorithm design and code quality.
- Lead undergraduate laboratory sessions, guiding hands-on exercises and reinforcing core concepts.
- Hold weekly office hours, providing individualized support to help students debug code, understand concepts.

Master's Teaching Assistant, Department of Computer and Information Sciences, Towson University Aug 2024 – May 2025

- Analyze curriculum data from 428 universities to evaluate NIST standard adoption, supporting new course development.
- Guide 60+ undergraduates per semester in lab sessions to reinforce programming concepts and support learning outcomes.
- Assist students with debugging and problem-solving to improve lab assignment completion and technical confidence.

Lab Administrator, Department of Computer and Information Sciences, Towson University Aug 2023 – May 2025

- Provide technical support to 1,700+ students and faculty by resolving software, hardware, and network issues.
- Assist faculty by troubleshooting technical problems in labs and classrooms, ensuring uninterrupted instruction.
- Configure lab environments through software installation and PC setup to streamline academic workflows.

Publications

C=Conference, J=Journal

[C.2] Krishna Sruthi Velaga, Anik Mallik, and Yifan Guo, "How Energy is Consumed by LLM-enabled Smart Home Assistant Systems on Low-Cost Devices: An Empirical Study", *Proceedings of the IEEE Consumer Communications & Networking Conference (CCNC)*, 2026. [Paper]

[J.1] Krishna Sruthi Velaga, Yifan Guo, and Wei Yu, "Edge AI for Smart Cities: Foundation, Challenges, and Opportunities" *Smart Cities*, 2025 [Paper]

[C.1] Krishna Sruthi Velaga, Yifan Guo "Optimizing Large Language Models Assisted Smart Home Assistant Systems at the Edge: An Empirical Study", AI4WCN AAAI 2025 [Demo] [Paper]

Education

PhD in Information Technology, Computer Science Track Aug 2025 – Dec 2028
Towson University, Towson, MD
Advisor: Dr. Yifan Guo
GPA: 4.00/4.00

Master of Science, Computer Science, Data Science Track Aug 2023 – May 2025
Towson University, Towson, MD
GPA: 3.97/4.00

Bachelor of Technology, Computer Science and Engineering Aug 2019 – Jun 2023
Vasireddy Venkatadri Institute of Technology (VVIT), Guntur, AP, India
GPA: 3.63/4.00

Projects

LLM-enabled Smart Home Assistant Systems on Low-Cost Devices: An Empirical Study [Paper] [Demo]
Tools: Python, LoRA, PyTorch, Raspberry Pi 4B/5, Home Assistant, HV Power Monitor

- Built a testbed with quantized LLMs on Raspberry Pi 4/5 with Home Assistant and thermal/power instrumentation.
- Collected multi-source raw data across 4 configurations accounting including log files, thermal readings, and power data.
- Synchronized heterogeneous timestamps to get fine-grained insights into performance, energy use, and thermal trade-offs.

- Reduced on-device response time by 82% (from 45.1s to 7.9s) through entity optimization, enabling faster responses.
- Improved intent recognition by fine-tuning TinyHome, TinyHome-Qwen, and StableHome models on synthetic datasets.
- Profiled workload distribution and identified inference as the dominant contributor ($> 80\%$) to total energy consumption.
- Conducted thermal analysis; Raspberry Pi 5 showed a 44% lower temperature rise, supporting stable sustained workloads.
- Evaluated battery performance with 3,000–5,000 mAh packs at multiple health levels: Pi 4B provided longer battery life but slower responses, while Pi 5 delivered faster responses but shorter battery life.

Automated Degree Completion Plan Assistant
Tools: Python, Qwen2.5(7B), Jina AI, Open WebUI, Ollama, Docker

- Scraped ∼300 academic webpages and converted course and program data into structured JSON, updated every 5 minutes.
- Built Open WebUI pipelines to route user queries (course lookup, plan generation, verification) using custom logic and retrieve relevant information from locally stored university webpage data.
- Generated semester-wise degree plans based on user input, prerequisite constraints, and basic credit-load considerations.
- Deployed system locally using Docker and Ollama with the Qwen2.5 model, avoiding external data transfer.

911 Call Analysis of Detroit and New York City [Code]
Tools: Python, TensorFlow, Matplotlib, NumPy, Pandas, Scikit-learn, Statsmodels

- Analyzed 7M+ 911 call data from Detroit and NYC to uncover patterns in call volume, response time, and incident types.
- Built predictive models (SARIMAX, XGBoost, DBSCAN, K-Means) achieving ∼85% accuracy in incident forecasting.
- Applied inferential statistics and clustering to identify peak crime times and delays in emergency response. allocation.

Predicting Ride Fares with Reinforcement Learning
Tools: Python, NumPy, Pandas, Scikit-learn, Matplotlib

- Processed NYC Taxi trip records with feature engineering (distance, time, passenger count, weather, and location data).
- Leveraged Q-Learning, DQN to predict optimal ride fares, formulating fare estimation as a state–action–reward problem.
- Visualized spatial and temporal pricing trends, revealing surge patterns in high-demand zones and times.

Malicious URL Detection using Machine Learning [Code]
Tools: Python, BeautifulSoup4, GoogleSearch-Python, Scikit-learn

- Developed a Chromium-based brower extension to alert users about malicious websites, enhancing online safety.
- Engineered 24 handcrafted URL features across lexical, host-based, and content-based categories for model training.
- Increased classification accuracy from 91.8% to 97.3% by integrating content-based features in the Random forest model.

Skills

Languages: Python, Java, SQL, C++, Bash, LaTeX
Libraries: Scikit-learn, TensorFlow, PyTorch, Keras, Pandas, NumPy, Matplotlib, Seaborn
Edge Computing: Raspberry Pi 4/5, Power Monitoring Devices
Technologies: Docker, Git, Home Assistant, Virtual Machines, Ollama, Open WebUI

Leadership

Vice Chairperson, Association for Computing Machinery Student Chapter, VVIT

- Collaborated with chapter officers to organize workshops, seminars, and other events weekly to enhance member engagement
- Managed event logistics during the VIVA-VVIT 2021 Annual Fest, ensuring smooth operations and successful outcomes.

Volunteering

Volunteer Computing, Berkeley Open Infrastructure for Network Computing (BOINC) Dec 2025 – Present